

How I Perceive It: Human Memory-Augmented Analogical Reasoning for Machine Visual Interpretation

Zhuodi Cai

Digital Arts & Experimental Media
University of Washington
Seattle, WA, USA
zc2525@uw.edu

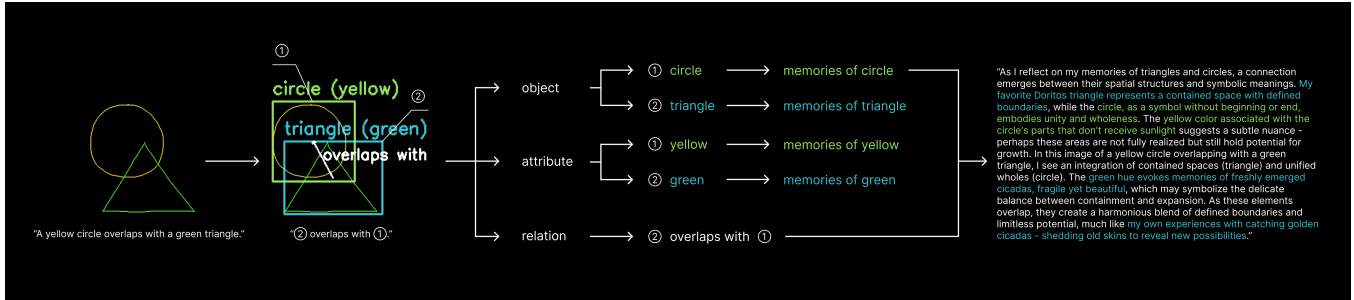


Figure 1: Brief overview of human memory-augmented analogical reasoning for machine visual interpretation process. An objective image caption is decomposed into two objects, two attributes, and one relation. Memory retrieval of objects and attributes enables analogical reasoning based on the relation, generating a new visual interpretation.

Abstract

How I Perceive It augments machine visual interpretation with human memory, shifting from superficial seeing to deep perceiving. Although today's vision language models (VLMs) can generate image captions with a degree of subjectivity, they still struggle to explain the underlying reasons or experiential basis for such subjectivity. Machines can see, but they do not perceive as humans do, who link perception with prior experience and memory. To bridge this gap, this paper introduces a visual interpretation system that integrates individual memory into machine perception, founded on structure-mapping theory. By merging what the machine sees with what the individual remembers, the system produces individualized interpretations that uncover more insightful meanings among visual elements that are not immediately visible on the surface.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Knowledge representation and reasoning*; • **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → *Media arts*.

Keywords

Human-AI Collaboration, Vision-Language Models, VLMs, Cognitive AI, Explainable AI, Visual Interpretation, Memory, Analogical Reasoning

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1 Introduction

Visual interpretation is a cognitive task shared by both humans and machines. Since the invention of the Perceptron [Rosenblatt 1958], machine learning models inspired by human brain and visual perception have evolved far beyond this linear classifier. Throughout the past 67 years, the field of computer vision has developed various new paradigms, algorithms, and applications. In terms of applications, vision models are now including object detection, image generation, depth estimation, pose estimation, image captioning, and semantic segmentation. VLMs, which fuse language models with vision models, facilitate advanced understanding of both still and moving images. For example, BLIP [Li et al. 2022] supports prompted image captioning, CogVLM2 [Hong et al. 2024] handles prompted video captioning, and ColPali [Faysse et al. 2025] performs document retrieval by recognizing visual information in documents.

Even so, machines' visual interpretation still differs from that of humans, due to differences in how both acquire experience and build memory. What machines can recognize does not necessarily align with human knowledge. In turn, when machines fail to meaningfully integrate recognized content, humans may interpret



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it relying on prior experience. As observed by researchers [Petridis and Chilton 2019], cultural background and familiarity with symbols could lead to misinterpretations of visual metaphors. They suggest that machine interpretations should not solely rely on annotated data, but also incorporate human knowledge as a necessary augmentation.

Therefore, *How I Perceive It* opens up a visual interpretation approach that selectively augments elements recognized by machines when they correspond to the individual’s memory archive, enabling interpretation augmented by human memory that is not offered by existing VLMs.

2 Background

2.1 Experience and Memory

One key aspect that distinguishes humans from machines is our experience and memory. Regarding experience, although machines can simulate human sensory experience, they still rely on different sensors and systems to process different signals. For visual information, cameras use electro-optical sensors to process electronic signals, while humans use photoreceptor cells to process bioelectrical signals. Memory is built on experience. *Thinking, Fast and Slow* explains that memory is a biased story told by the remembering self that reconstructs past events based on what the experiencing self lived through [Kahneman 2011].

2.2 Augmented Memory

However, assuming that machines could simulate human sensory input, humans still cannot, like machines, accurately and infinitely remember all data points from our perceptual systems. The selectivity of human memory contributes to its vagueness and allows us to think through forgetting. A fantasy story *Funes el memorioso*, or *Funes the Memorious*, connects thinking with the ability to forget or blur memories. Funes could only remember details and never forget the differences between them, making general thinking and abstraction impossible [Borges 1942].

But that does not imply machine memory and human memory are mutually exclusive. Machine’s data sensing and storage capabilities can supplement human memory, thus creating the augmented human, a concept already envisioned in science fiction. For instance, *Six Wakes* imagines a mindmapping technology for modern cloning that records a person’s new experiences, memories, and emotional development within minutes [Lafferty 2017]. *A Memory Called Empire* proposes an imago-machine that replicates a person’s memories and cognitive patterns into an imago, which can decouple and couple to switch hosts, making it possible for humans to live beyond physical limits [Martine 2019]. Its sequel, *A Desolation Called Peace*, further indicates that the imago preserves personal identity continuity [Martine 2021].

2.3 Memory and Analogical Reasoning

Memory, as a source of familiar experiences and knowledge, provides the base domain for analogical reasoning, through which we form analogies and generate new ideas about target domains. According to *Visual Analogy: Consciousness as the Art of Connecting*, through intuition or beholding, we come to know what is innate,

and insight allows us to perceive the ontological and phenomenological similarities that connect seemingly unrelated structures on the surface [Stafford 1999]. *Structure-Mapping: A Theoretical Framework for Analogy* proposes that analogy interprets unfamiliar target domains by mapping relational correspondences from familiar base domains [Gentner 1983]. Rather than merely focusing on objects and attributes, the theory emphasizes interconnected relational structures.

For example, upon seeing a drawing of a flower with a head shaped like a cube, a viewer may activate associative pathways like cube→ice→cooling, cube→3D→rotation, cube→angularity→sharpness, and stem→plant→fragility. With these associations, the viewer treats the head as a geometrical, artificial object and the stem as a slender organic support. Thus, the structure-mapping may blend into an interpretation like “This flower has a rotating ice cube head that can be picked in the summer to cool down beverages, but one must be careful when picking due to the sharpness. Such a flower does not seem to exist in nature, perhaps it is from human imagination or an artificial species. A slender natural stem supports a massive human-made ice cube head, nature bears the weight of human ambition.”

From the perspective of cognitive neuroscience, with more detailed queries, information is more likely to be activated from memory, as scientists note in *Memory Retrieval and the Functional Organization of Frontal Cortex* [Nyhus and Badre 2015]. Focusing on “rotation”, for instance, might lead to associations with a spin around a fixed axis like the Earth’s rotation, or rotation based on relative positioning like a three-axis stabilized CubeSat, a type of nanosatellite. Section 3 describes the overall method, including how the information is selected and how memory retrieval is performed.

3 Method

To computationally model how human memory-augmented analogical reasoning may support machine visual interpretation, a memory archive is first created using the artist’s personal recollections and past journal entries. The archive consists of fragmented memories from the age of three to the present, each stored as a separate textual entry within a single text file. Examples include some that are factual, such as “Wood ear mushrooms grow on damp logs.” Some contain metaphors, like “That day, when he pretended, I saw the prosthesis of his soul.” Such metaphorical memories, already abstracted by humans, can vary even more between individuals than factual ones. Through augmentation, these memories, in particular, take machine understanding to a new dimension.

A six-step system is then developed to simulate this interpretation process. A visual overview of Steps 2 to 6 is provided in the teaser or Figure 1.

Step 1: Captioning. The audience uploads an image, for example, a drawing of a flower. Then a VLM describes the visual content in text. The raw caption includes both objective details like “elongated petals” and subjective descriptions like “visually appealing”.

Step 2: Removing Subjective Descriptions. The raw caption is subsequently processed by a large language model (LLM), which is prompted to remove any subjective content obtained from the VLM. For instance, the sentence “The image depicts a stylized flower with

a vibrant pink center and delicate, elongated petals” is cleaned up to “The image depicts a flower with a pink center and elongated petals.”

Step 3: Extracting Visual Elements. The cleaned caption from Step 2 is used to extract key visual elements including objects, attributes, and relations, which are later used to do analogical structure mapping. For example, “stem” is the object with an attribute of “blue” that has relations to other elements like “leaves located on stem”.

Step 4: Filtering Elements Based on Memory. For each element, initial filtering is performed through keyword matching, because with sufficiently large annotated datasets, a machine may recognize elements that do not belong to the individual’s base domain of familiarity. Objects and attributes are used to determine which ones are supported by memory. For example, if the element “leaves” from Step 3 appears in the archive, it suggests that the viewer has prior familiarity with the object. The corresponding memory entries are therefore preserved and passed on to Step 5.

Step 5: Retrieving Memories with Filtered Elements. The elements that pass the initial filtering are used to identify an initial set of relevant memories. These initial memories, together with all the others, are semantically embedded. Then, cosine similarity is calculated between the initial memories and the rest, and a top-k is applied to retrieve the most closely associated ones. Building on the recognition established in Step 4, this step simulates how visual input may activate not only directly related memories but also indirectly or associatively connected ones.

Step 6: Constructing Interpretations. The target domain is the uploaded image, while the base domain consists of objects and attributes recognized by both machine and human memory, along with interpretive memories retrieved through these elements via keyword matching and semantic similarity. Guided by relations, an analogical structure is built between the two domains, leading to new interpretations.

Considering system accessibility, a cross-platform web client is integrated alongside the visual interpretation component. The client communicates with a GPU server for accelerated data processing. Figure 2 shows the system, from human-machine interaction to machine perception.

4 Implementation and Results

Once an image is uploaded, the web client sends an HTTP POST request with the image data to the “/process” endpoint on the remote GPU server. A Docker container running a Flask application inside the server processes the image, performs visual interpretation, and returns the results back to the client, including images and text illustrating how human memory reframes machine perception.

On the server, VLM Moondream2 [vik 2024] generates a raw image caption with a query, then LLM Llama-3-8B-Instruct [Meta 2024] removes subjective content and keeps objective descriptions, which are structurally decomposed into objects, attributes, and relations with support from spaCy’s `en_core_web_sm` [Explosion 2024] natural language processing pipeline. Figure 3 shows six visualized examples for the current phase.

To make the keywords used in the initial memory filtering more flexible, variants of element words are generated before keyword matching, for example, generating “flowers” from “flower”. After obtaining the keywords and their variants, memory matching is performed. The text encoder `nomic-embed-text-v1.5` [Nussbaum et al. 2024] semantically embeds both the initially filtered memories and the entire memory archive and then calculates similarities to retrieve relevant ones. After obtaining the matched and relevant memories, the set of memories corresponding to the objects and their attributes, the relations between the objects, and the objective descriptions together are used as the prompt for the LLM. This ensures that the model not only incorporates memories directly into a new interpretation, but also emphasizes constructing explanations based on the relations, in accordance with the structure-mapping theory of analogical reasoning. The final visual interpretation augmented by human memory, as an instance, is shown in Figure 4.

5 Application

5.1 Art Practice

As a minimum viable system, *How I Perceive It* has been incorporated into an ongoing art project. With this core framework as a foundation, custom modules are developed to support broader experimental practices.

One extended version generates each object’s image based on retrieved memories. Through the conditioned visual generation of each object prompted by associated memory descriptions, this practice creates a collage artwork that asks “How might human memory enrich what machines have seen? And what new meanings could it convey?” For each recognized object, a corresponding image is generated and composited with the others into an animation. The result overlays machine-recognized content with visuals prompted by memories, and applies glitch effects with the help of a JavaScript library [7PH 2025] to reflect the inaccuracy and fragmentation of human memory, as well as the ambiguity in the visuals reimagined by the machine. The animation is then combined with the recomposed glitch sound effects, originally generated by MusicGen [Copet et al. 2023], to create a video. Some example frames are shown in Figure 5.

Instead of presenting the interpretation as text, the artist uses a voice cloning model to narrate the result in her own synthetic voice, reinforcing the fusion of memory and machine into a continuous self. To generate objects’ images, the rectangular regions (x_0, y_0, x_1, y_1) containing each detected object are processed separately. Canny edges are extracted using the OpenCV library [Bradski 2000], and segmentation masks are obtained via Segment Anything Model [Kirillov et al. 2023]. These are used to condition a Stable Diffusion model [Rombach et al. 2022] via ControlNet [Zhang et al. 2023], with scales of 0.2 and 1.0 respectively. This process is illustrated in Figure 6.

Figure 7 shows the setup and files used in the installation. After the audience draws and uploads the image to the remote server, *How I Perceive It* and custom modules start processing it. The three drawings in the first row of Figure 3 are examples. Once the results are sent back to the client, the code recompiles and reads the new files. This produces the final audiovisual result displayed on the large screen. The interaction time between the audience and the

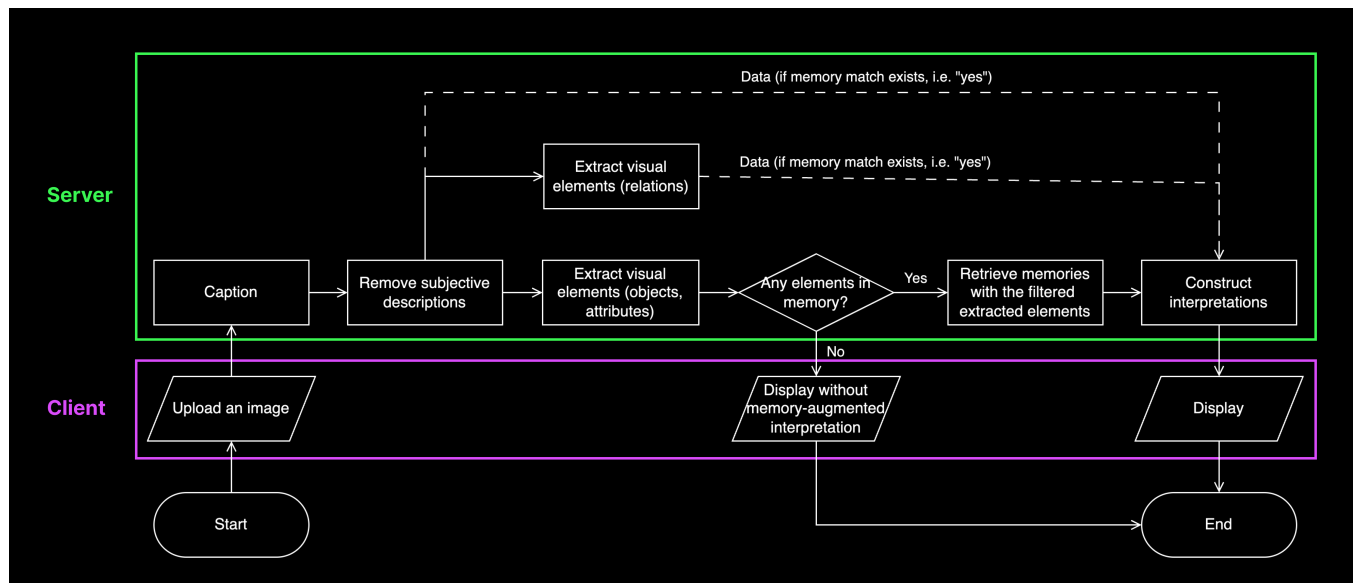


Figure 2: System overview from image upload to memory-augmented visual interpretation display. In the figure, two colors are used to differentiate the server and client components. If there is no memory containing the elements, display without memory-augmented interpretation and then end.

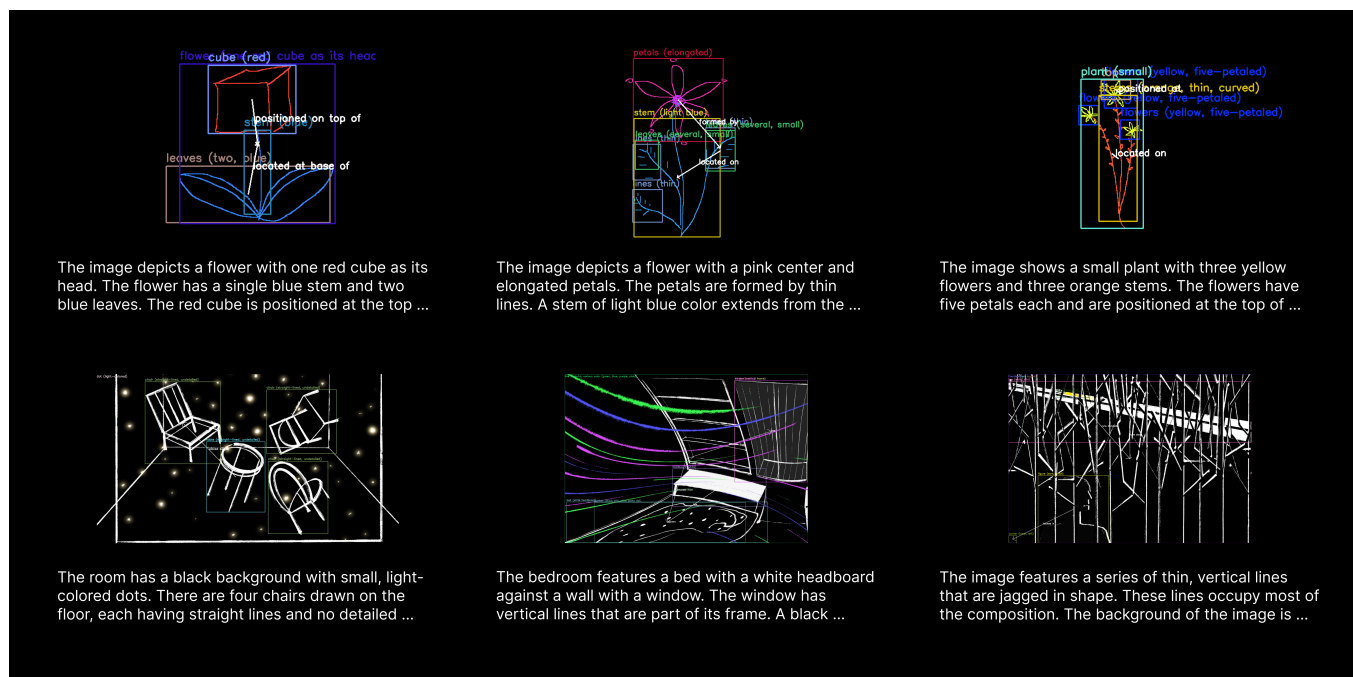


Figure 3: Objective caption and structured visual representation from objective caption. The white text below the image is from the objective captions processed from the raw captions. The image is based on the original, with added information extracted from the objective captions, including objects, attributes, and relations. Specifically, the involved objects are marked with colored bounding boxes and object names. The attributes of objects are written in parentheses next to the object name in a uniform color. White arrows and text are used to explain the relations between the objects.

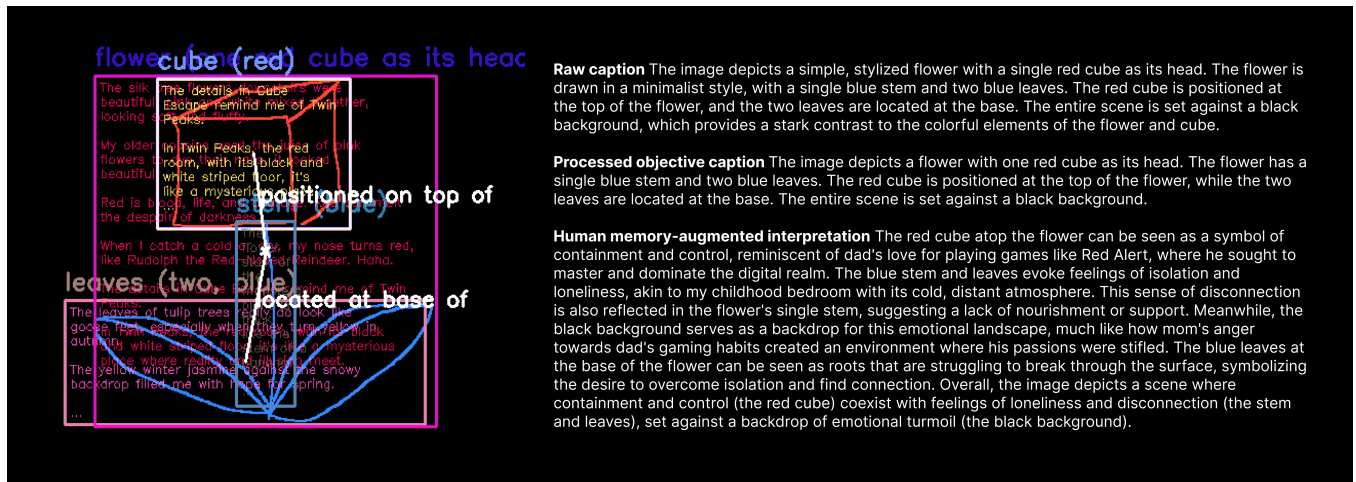


Figure 4: Before and after interpretation augmentation. The colorful image is a composite created by overlaying several images. Except for the original image and the visualized objects, attributes, and relations extracted from the objective caption, it also includes textual memories retrieved based on each object and its attributes, which are displayed inside the objects' bounding boxes. The white text presents the three main stages and results of the machine visual interpretation process, which are raw caption, processed objective caption, and human memory-augmented interpretation.

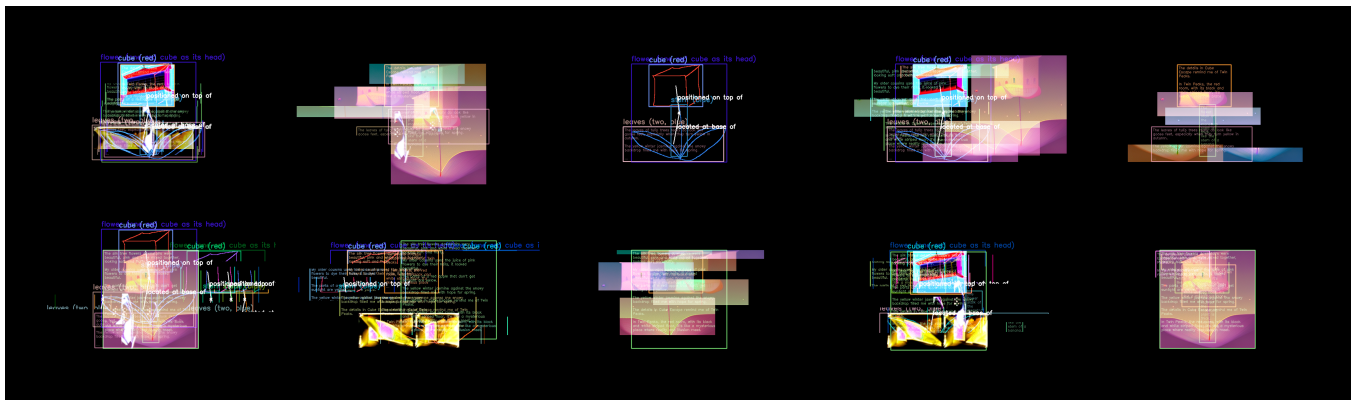


Figure 5: Example frames. Images are displayed randomly, accompanied by randomized glitch effects.

installation depends on the duration of the audience's operation, image size, and image complexity.

A screen recording of the animation is included in the supplemental materials. The red cube flower with a blue stem and two blue leaves, introduced earlier, is used here again to ensure consistency.

5.2 Introspective Practice

Another potential application lies in introspective practice that fosters self-awareness and self-healing.

For instance, for a decade, the artist had recurring nightmares involving stairs. These stairs had attributes like "broken" and relations to other objects like "an abyss is at the bottom of the stairs". By retrieving memories related to "stairs", "broken", and "abyss", and then interpreting the scene, she found the emotional roots of fear. Through creating positive experiences and consistent practice, she gradually healed. Over two years, she applied this process to other

nightmares, which transformed her dream narratives. Since then, she has no longer experienced nightmares, because this method allows her to intervene early in potentially negative dreams, preventing them from escalating. As part of this process, the artist visualized these ephemeral moments in dreams by sketching them after waking up, and three of them are shown in the second row of Figure 3.

How I Perceive It can assist this process by showing how the self might interpret an image under the influence of particular memories. Much like how one might attempt to interpret a dream, it generates a subjective visual interpretation that encourages one to critically question "Is this really how I perceive it? And why?" By exposing memories through memory-augmented machine interpretation, the system provides a mirror that invites people navigate and discover psychological patterns.

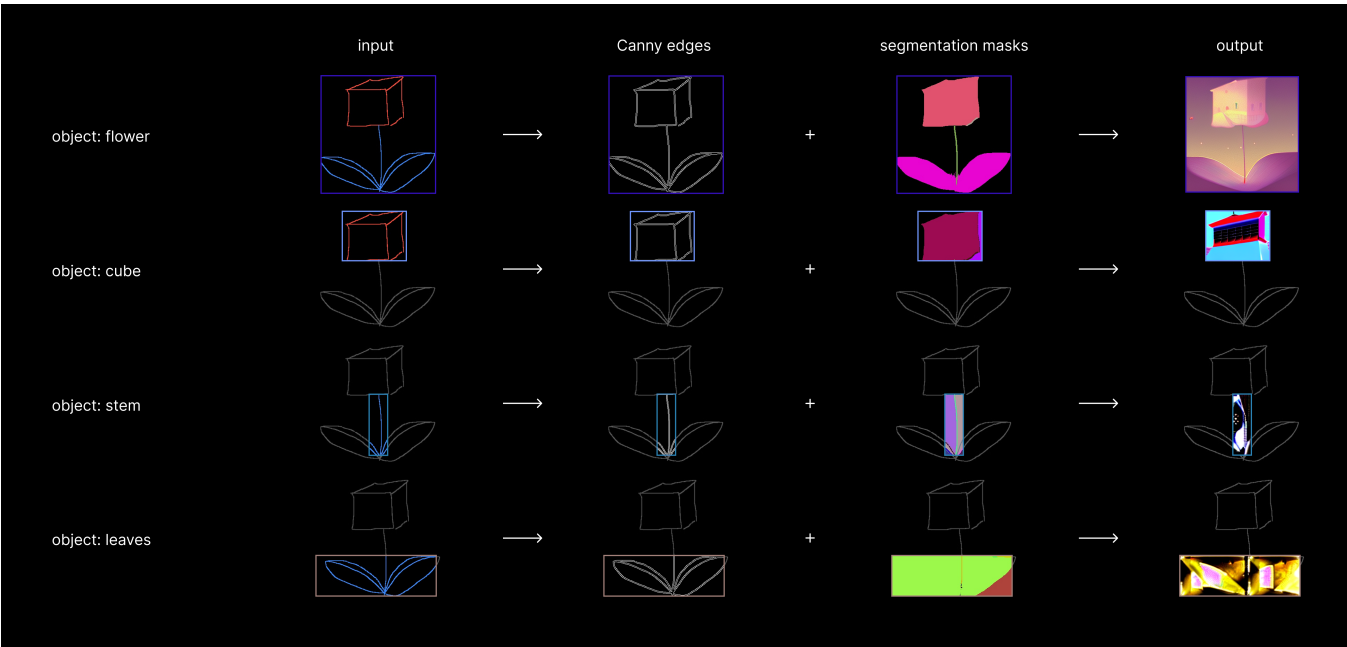


Figure 6: Image processing and generation workflow. The workflow shows the process from input images to Canny edges and segmentation masks to output images. The light gray part is only used to assist in illustrating the position of the object's image in the original image, and is not involved in the Canny edge detection, image segmentation, and image generation steps.

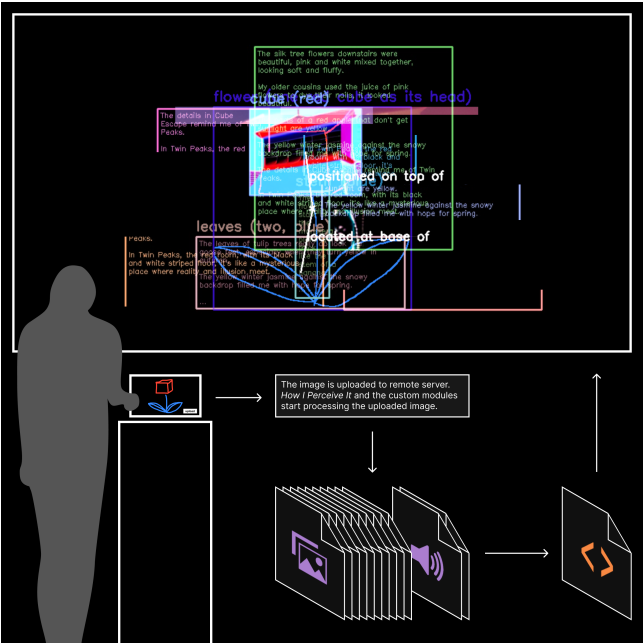


Figure 7: The setup of the installation and the materials it uses. In the example, the materials include ten images, two sound files, and one code file that can be recompiled when new audiovisual materials are received from the server.

5.3 Communication Enhancement

How I Perceive It can also be applied to visual communication scenarios, such as intercultural communication and marketing communication.

In the context of intercultural communication, the same visual symbols can be interpreted differently by people from different cultural backgrounds. For example, the magpie is often considered a symbol of good luck and auspiciousness in Eastern cultures, whereas in Western countries it has higher probability to be interpreted as an ill omen. However, despite the differences in how magpies are perceived in Eastern and Western cultures, they can still be associated with a common symbolic meaning, the conveying of omens. Magpies have generally been seen as symbols of good omens in China since ancient times. For example, in *Mo Yu'er*, a poem from the Southern Song dynasty by Zhao Shankuo, the magpie is explicitly described as a foreknowing messenger. Similarly, in Ancient Rome, the modern word “auspicious”, meaning good fortune, comes from the Latin “auspiciu”, which refers to interpreting omens through bird observation, with the “auspex” being the person in charge. As a bird, the magpie may have been included among those observed for omens.

Thus, if marketing communication has intercultural communication purpose, it is necessary to consider the audiences’ knowledge background to ensure the visual content is not misinterpreted. In this regard, the system can utilize collective memories rather than just individual memories to help simulate, evaluate, examine, and enhance the effectiveness of visual communication.

6 Discussion

6.1 Model Limitations

The system's performance is limited by the models' capabilities. For instance, the current VLM sometimes has difficulty accurately recognizing all objects in an image, let alone their attributes and relations. Therefore, improving the recognition ability of these models is crucial, though this may be limited by human cognition itself, making them suitable only for human memory-augmented visual interpretation. For example, if the memory archive belongs to N6MAA10816, Roy Batty the replicant in *Blade Runner* [Scott 1982], his unique memory of C-beams glittering in the dark near the Tannhäuser Gate would participate in the interpretation process. However, the models trained on data collected by humans may have never seen C-beams nor the Tannhäuser Gate, and the captions would not provide corresponding objects. In this case, memories directly related to these elements would initially fail to match and would thus rely on the subsequent semantic search.

6.2 Memory Completeness

The performance is also constrained by the quantity and quality of the human memories utilized. Insufficiently detailed memory limits the comprehensiveness of retrieval. This challenge relates to whether cognitive neuroscience methods can be effectively adopted to collect detailed memories and whether individuals are willing to fully disclose all details. In addition, text is not the only medium for memory, other media should be considered in the future.

6.3 Perceptual Preferences

But even with the most powerful models and abundant individual memories, this system still has not taken into account each individual's visual perceptual habits and cognitive patterns.

People tend to focus on different visual elements, memories, and emotions. The order and weighting of visual element extraction, as well as memory retrieval habits, can vary. Some people may first notice attributes like color, while others may first recognize objects. Some may recall sad memories, while others recall happy ones. That is, if a machine could learn an individual's way of visual perception and cognitive patterns, and adapt accordingly, this human-machine collaboration could be more interpretable and explainable. By collecting and analyzing temporally aligned multimodal data, the machine may discover the relationship between experience and memory, thereby gaining insight into an individual's perceptual preferences. For example, when an individual begins looking at an image, an eye-tracking system could start collecting data as the person describes the visual information and the memories it triggers.

6.4 Evolving Self

As human experiences and memories constantly grow, our ways of perceiving may change over time. For this reason, timely, if not real-time, data collection and analysis are essential for an evolving machine, which also reflects the necessity for the fictional mindmapping technology in *Six Wakes* to record new experiences, memories, and emotional development within minutes rather than hours or longer.

By integrating both the experiencing self and the remembering self, *How I Perceive It* would transcend mere interpretation and become an extension of the self or even the self, as the imago is imagined in *A Memory Called Empire*.

7 Conclusion

Through investigating the interconnections among experience, memory, and analogy, *How I Perceive It* explores the differences and collaboration potentials between humans and machines in visual interpretation. Based on the theory of analogical reasoning, it proposes and implements a method to augment machine visual interpretation by filtering and retrieving human memories through structured information. It further identifies three directions of application: art practice, introspective practice, and the enhancement of communication in intercultural and marketing contexts. As machines adapt to individual perceptual habits and preferred memory media, a human-machine collaboration like *How I Perceive It* would enable deeper understanding in real-time according to the latest experiences, memories, and cognitive patterns, offering more insightful interpretation.

Like Funes, a machine may remember too much to abstract meaningfully. Meaning may emerge not from what could be remembered, but from what must be forgotten. Through entanglement, human and machine jointly decide what to discard and what to preserve, thereby constructing meaning.

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